

WERNIGERODE, Germany

WMO: 104540

Lat: 51.8456N

Lon: 10.7686E

Elev: 769

StdP: 14.29

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.4	12.4	0.4	5.8	9.2	5.9	7.6	14.0	32.3	43.7	29.5	41.5	2.5	330	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	82.4	65.2	79.3	63.9	76.2	62.4	66.9	78.9	65.2	76.5	63.7	74.1	8.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	87.2	72.2	60.7	81.4	70.2	59.0	76.7	69.2	31.8	78.0	30.5	76.5	29.3	74.3	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	23.8	20.3	DB	2.7	89.2	10.2	2.8	-4.6	91.2	-10.5	92.8	-16.2	94.4	-23.6	96.4	(4)
(5)			WB	2.8	70.6	9.9	2.9	-4.4	72.6	-10.2	74.3	-15.8	75.9	-23.0	78.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	33.5	31.1	38.8	45.1	53.5	58.8	63.1	62.9	56.9	49.5	40.3	36.3		
	DBStd	13.23	9.98	10.59	7.75	7.06	6.17	5.94	5.67	5.56	5.31	6.76	7.63	8.12		
	HDD50	2434	511	530	352	180	36	3	0	0	5	91	298	428		
	HDD65	6473	976	948	813	598	359	202	106	108	250	483	741	889		
	CDD50	1549	0	1	4	32	144	267	406	399	211	74	7	4		
	CDD65	114	0	0	0	0	1	16	47	43	6	1	0	0		
	CDH74	1153	0	0	0	1	37	174	469	408	58	6	0	0		
	CDH80	217	0	0	0	0	2	31	92	91	1	0	0	0		
(14) Wind	WSAvg	7.5	9.9	7.9	8.4	7.3	6.2	6.4	6.6	6.2	7.2	7.4	8.0	9.1		
(15) Precipitation	PrecAvg	23.1	1.7	1.3	1.7	2.0	2.6	2.7	2.0	2.4	1.3	1.6	2.0	1.8		
	PrecMax	29.6	5.3	3.4	3.8	5.3	4.9	6.0	4.8	3.9	2.5	3.8	3.6	4.8		
	PrecMin	16.8	0.2	0.2	0.4	0.6	1.3	0.6	0.8	1.2	0.3	0.3	0.9	0.1		
	PrecStd	4.6	1.1	0.9	1.1	1.2	1.1	1.4	1.0	0.9	0.6	1.0	0.9	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	58.4	63.0	70.8	77.9	84.2	86.1	87.8	79.5	74.2	61.0	56.7		
		MCWB	48.0	47.8	49.8	54.9	60.7	66.3	66.7	66.8	62.3	60.4	52.5	47.3		
	2%	DB	49.7	52.7	56.7	65.4	74.1	79.0	82.5	82.0	75.2	67.7	56.4	53.5		
		MCWB	44.8	45.4	47.9	52.6	58.7	64.5	65.7	64.3	61.4	57.4	49.8	47.1		
	5%	DB	47.5	48.0	52.6	61.1	70.2	75.0	79.3	78.5	70.9	63.2	53.4	50.8		
		MCWB	43.1	42.8	46.0	50.0	56.7	61.7	64.4	63.6	60.0	55.3	48.4	45.7		
	10%	DB	45.2	44.6	49.4	57.2	66.5	71.0	75.8	75.0	66.8	59.8	50.9	48.0		
		MCWB	40.9	40.3	44.0	48.3	55.6	60.4	63.0	61.8	57.7	52.9	46.5	43.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	49.3	51.3	56.6	63.5	69.6	69.7	68.4	64.8	61.8	53.2	49.6		
		MCDB	52.5	57.3	59.2	67.9	73.2	79.2	81.7	83.2	75.7	72.1	57.1	55.0		
	2%	WB	45.8	46.1	48.8	53.7	60.5	65.6	67.4	65.9	62.7	58.4	50.5	47.7		
		MCDB	49.5	51.9	56.4	63.8	70.0	75.9	79.0	78.8	72.5	65.8	55.3	52.6		
	5%	WB	43.4	43.1	46.5	50.9	58.3	63.3	65.6	64.5	60.5	55.7	48.7	45.9		
		MCDB	47.0	47.9	51.9	59.3	68.0	72.7	76.7	76.0	69.1	62.2	53.1	50.4		
	10%	WB	41.1	40.5	44.3	48.8	56.1	61.1	63.9	62.8	58.6	53.6	46.7	43.6		
		MCDB	44.9	44.5	49.3	56.7	65.6	70.0	74.5	72.9	66.7	59.2	50.9	47.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.5	12.2	15.0	17.5	16.0	17.3	17.5	15.1	13.5	10.6	9.4		
		MCDBR	12.1	15.2	18.2	20.7	24.9	23.6	24.3	23.8	22.8	20.0	14.9	11.3		
		MCWBR	10.1	11.5	12.4	12.2	13.7	12.6	12.1	11.5	12.6	13.1	11.0	9.0		
	5% WB	MCDBR	10.6	15.2	16.6	18.6	21.4	20.5	22.1	21.2	19.6	18.2	13.7	10.3		
		MCWBR	9.7	12.0	11.9	12.0	13.1	11.9	11.8	11.2	11.7	12.6	10.3	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.334	0.373	0.411	0.414	0.410	0.427	0.420	0.382	0.354	0.332	0.305		
	taud		2.369	2.340	2.251	2.196	2.222	2.256	2.215	2.253	2.347	2.410	2.400	2.368		
	Ebn at Noon		220	247	257	261	266	268	260	256	252	237	209	201		
	Edh at Noon		19	26	35	41	42	41	42	39	31	24	19	16		
(44) All-Sky Solar Radiation	RadAvg		222	443	784	1269	1556	1680	1577	1345	979	562	258	169		
	RadStd		25	61	89	152	180	138	213	122	120	61	31	18		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page